GDP360 Hackathon – Solution Document Template

Track: Data Analytics & Business Intelligence – Macroeconomic Forecasting with ETL, SQL & Power BI

---

1. Student Information

Field	Details
Full Name	Priyansh nandwana
Registration Number	RA2211003010886
Department / Branch	CTECH/CSE core
Year	4th
Email ID	pn6846@srmist.edu.in

---

2. Problem Scope and Track Details

Section	Details
Hackathon Track	GDP360 – Data Analytics & BI
Tools Used	✅ SQL ✅ Alteryx(Python since i have mac) ✅ Power BI(Tableau, same reason)
Core Modules Implemented	✅ (Check all that apply)
	✅ Data Ingestion & Cleaning (Alteryx)
	✅ Master Table SQL Transformation
	✅ Macro Indicator Analysis
	✅ Country Growth Dashboard (Power BI)
	✅ Global Coupling Analysis
	✅ Sector-Wise Trend Analysis

Optional Enhancements / Scenarios Implemented

- Scenario 1: *GDP Anomaly Detection using 3-sigma statistical method*
- Scenario 2: *Top 20 economies identification with year-wise comparison*
- Scenario 3: *Interactive Region- and Country-Level Dashboard with Filters*

---

3. Architecture & Design Overview

- **Data Pipeline Diagram**
  (Attach diagram showing data flow: ingestion → transformation → analysis → visualization)
- **SQL Schema Design**
  (Show entity relationships like Country ↔ Year ↔ GDP Metrics)
- **Overview of Tools Used & Rationale**
  - **Python (Pandas)**: For all data ingestion, transformation, anomaly detection, and export to final master_data.csv
  - **SQLite**: Lightweight, fast relational DB to store and run analytics using SQL views
  - **Tableau**: BI tool of choice for clean, interactive, multi-page dashboards

---

4. Feature-wise Implementation Details

Repeat the following for each core feature:

Feature: GDP Anomaly Detection

- **Scenario Brief**
  Identify unusual spikes in a country's GDP, indicating major events like economic reforms, oil discoveries, or recovery.
- **Key Tasks Performed**
  - Grouped GDP data by country
  - Calculated mean and std deviation per country
  - Flagged years with GDP > $\mu + 3\sigma$

- **Sample SQL Queries / Logic**

```
SELECT * FROM gdp_data
WHERE GDP_Anomaly_Flag = 1;
```

- **Outputs & Screenshots**
  Detected 8 anomalies (e.g., Guyana 2023-24, Ethiopia 2022)

- **Challenges Faced & Resolution**
  Ensured countries had enough data before applying 3σ; filtered invalid entries

Feature: YoY GDP Growth Calculation

- **Scenario Brief**
  ITrack yearly GDP progression to identify growth, stagnation, or recession
- **Key Tasks Performed**
  - Used LAG() to compare current and previous GDP
  - Rounded % growth to 2 decimal places
- **Sample SQL Queries / Logic**

```
SELECT Country, Year, GDP,
    ROUND(((GDP - LAG(GDP) OVER (PARTITION BY Country ORDER BY Year)) * 100.0) /
        LAG(GDP) OVER (PARTITION BY Country ORDER BY Year), 2) AS YoY_Growth_%
FROM gdp_data;
```

- **Outputs & Screenshots**
  View used in line charts and tooltip summaries

- **Challenges Faced & Resolution**
  Managed nulls when prior-year GDP was missing

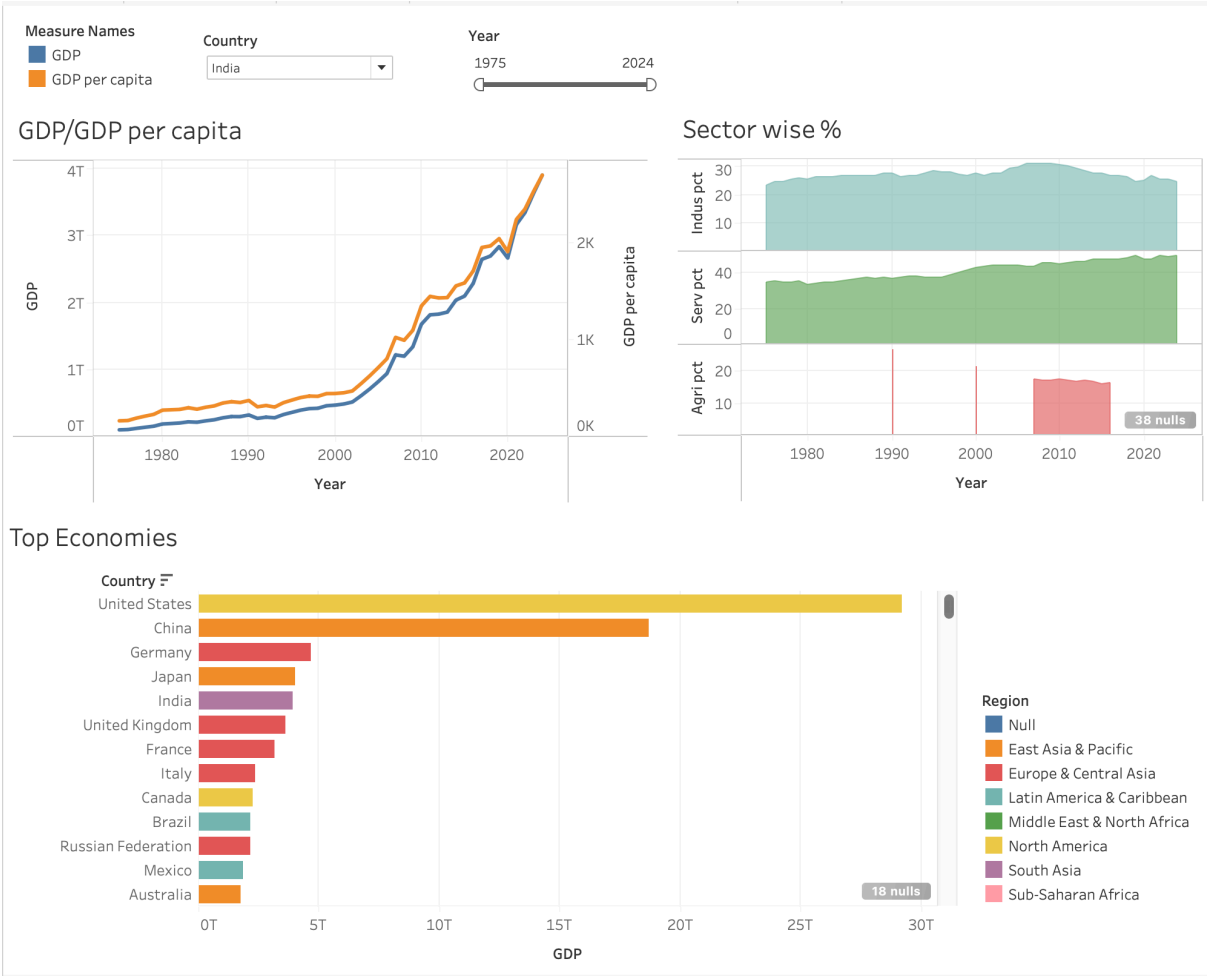
Feature: Tableau Country Dashboard

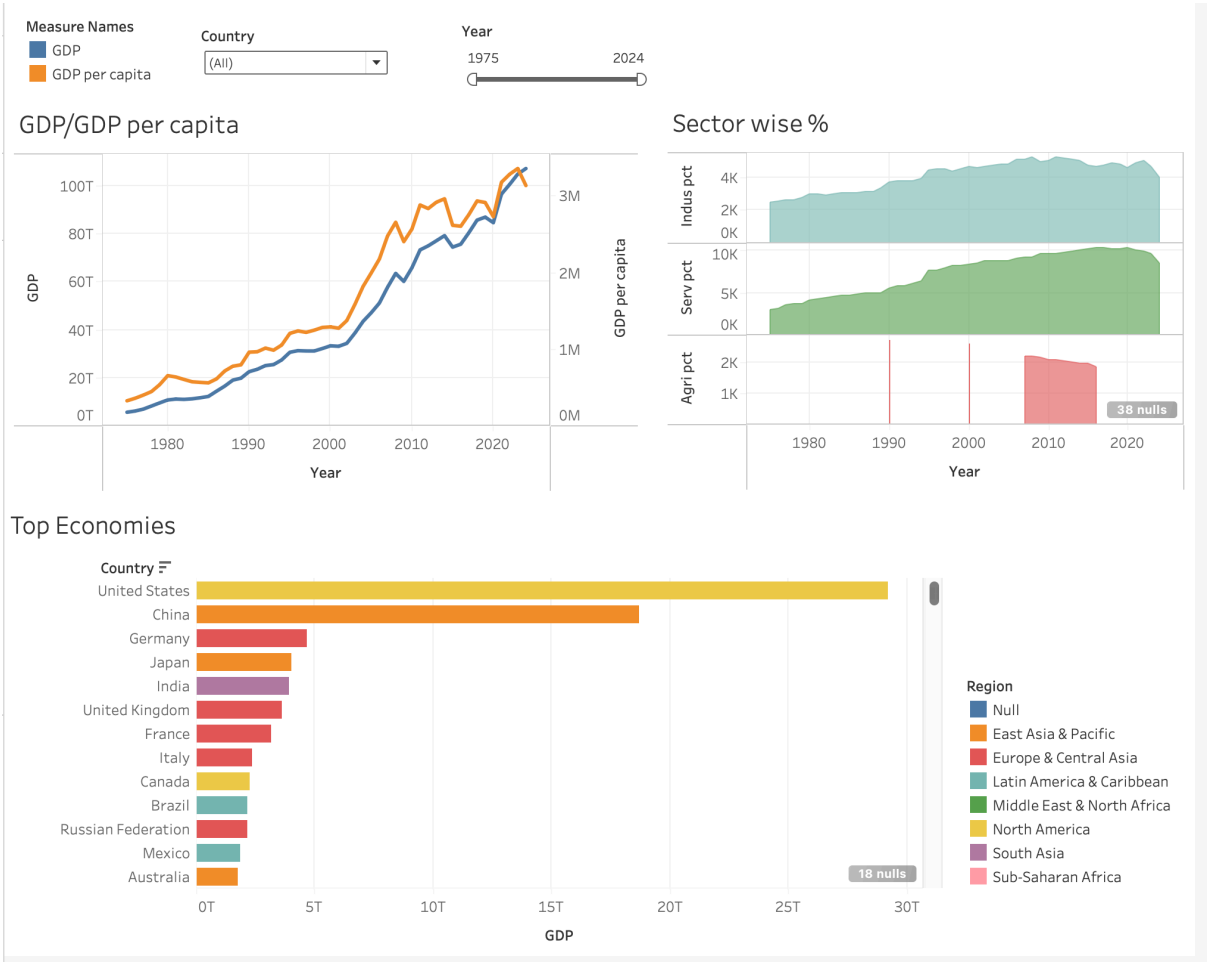
- **Scenario Brief**
  Visualize and compare GDP, population, inflation, unemployment, and sector trends for selected countries

- **Key Tasks Performed**
  - Created filters for country and year
  - Used stacked area charts for sector trends
  - Used scatter plots for inflation vs unemployment

- **Outputs & Screenshots**
  7 Tableau dashboard pages

  Interactive slicers and drop-downs





- **Challenges Faced & Resolution**
  Converting data types, ensuring consistent formatting for Tableau input

---

5. Testing & Validation

Category	Details
Number of SQL test queries run	25+ queries tested including joins, filters, aggregates
Number of edge scenarios handled	Countries with missing years, zero GDPs, incomplete sector data
Visual & Interactive Features tested	Filters (Country, Year, Region), Hover Tooltips, Dual Axis for GDP & per capita

Known bugs / unresolved limitations	Sparse agriculture data (~22% coverage); Minor inconsistencies in unemployment after 2020

---

6. Final Reflection & Submission

- **Key Learnings from this Hackathon**
  Built a complete **ETL + SQL + Tableau** pipeline from scratch

  Learned how to handle **messy economic data**, reshape wide/long datasets

  Deep dive into **3-sigma statistical outlier detection**

- **Strengths of your Implementation**
  Modular and commented Python ETL pipeline

  Reusable and optimized SQL views

  Clean, interactive, multi-level Tableau dashboard

- **What You Would Improve**
  Include **predictive analytics or forecasting**

  Fill gaps in agriculture/unemployment data

  Add **comparative rankings and KPIs** in dashboard

- **Career Alignment**
  Analyst, Data Science, Business Analyst